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Finding Suitable Houseplants

Questions

1. What plants would be best suited for the Seattle area?
2. How does the annual percentage of sunshine affect temperature fluctuation and precipitation in different cities?
3. How do environmental conditions such as temperature and humidity influence plant health and growth indoors?

Motivation

Adding plants into one's household not only creates a better ambience, but also offers a rewarding hobby in maintaining the plants. However, figuring out which plants would be best suited for a space may become difficult and expensive. Understanding the relationship between environmental conditions (temperature, sunshine) and plant growth is essential for picking plants that will survive in different cities.

Datasets we will use

- Our first dataset we're using is the "Indoor Plant Health & Growth Dataset" made by the user Souvikrana17, which contains data of different types of indoor house plants, including their height, leaf count, watering trends, environment conditions, and a health score.
<https://www.kaggle.com/datasets/souvikrana17/indoor-plant-health-and-growth-dataset>
- The second dataset we're using is the "Annual percent of possible sunshine by US City" dataset made by the user thedevastator, which contains the monthly average possible sunshine per month for 159 U.S. cities.
<https://www.kaggle.com/datasets/thedevastator/annual-percent-of-possible-sunshine-by-us-city>
- The third dataset we're using is the PRISM Weather Data by Zipcode, County, or State, which uses data collected from the PRISM Climate Group in Oregon State University, created by Seunghyn Lee. Like the title states, this dataset provides temperature and precipitation data for different areas in the contiguous United States from 01/01/1981 to the present. In particular, we are using the 2025 weather dataset based on State, which contains monthly temperature and rainfall data for most cities in the U.S.
<https://www.aaronsmithagecon.com/download-us-weather-data>

Method

Determining the machine learning models

To determine the machine learning model used for predicting the health scores, we considered the amount of parameters being used to train the model. Since we had multiple plant entries and multiple plants with different climate parameters, we decided to utilize the XG gradient boosting classifier model, which implements gradient boosted decision trees. It works by combining weaker models to create a stronger learner and correct the mistakes made by the previous ones, which works well with our project as it increases the accuracy of the model. It also implements L1 and L2 regularization to prevent overfitting of the datasets. To decide the right multiple linear regression model used to determine correlations of multiple predictors and the plant health score, we decided on using the OLS linear regression from the statsmodel.api as it provided convenient methods for collecting the coefficients of each predictor to determine their correlations with the health score.

Before every question, we had to do preprocessing of each data set, which included dropping unnecessary/unused data and merging the sunshine and temperature/precipitation dataset to create an overall city climate dataset. Since the temperature/precipitation dataset only covered the overall state's temperature, to merge the two datasets, we obtained the state codes from each city entry in the city sunshine dataset and merged them on that value. Furthermore, to make the data analysis easier, we expanded the merged data to have a row entry for each city month for a total of 12 entries per city.

Q1: - Retrieve plant health and growth data

To answer this question, we trained a machine learning model to predict a given plant's health score with the predictors of temperature and sunlight. For this particular problem, we utilized an XGBoost classifier model as it handles multiple, non-linear data well. We also needed to format the city data in a way that was similar to the plant data so that the model could predict the score of each plant in a given city. To do this, we condensed each monthly city data entry to create an annual city climate profile with the average sunlight and temperature, along with adding boolean entries for the variability of a city's weather and sunlight over the year. To determine the variability, we checked if the standard deviation of each climate condition was higher than a determined threshold (sourced from the Internet to find reasonable values). We then created a copy of the given city's climate, renaming the columns to match the plant data structure, and used the model to predict what the health score would be for each plant if it were in the given city's climate. To find the most suitable plants for the city, we added the plants to a list of best plants only if their predicted health score was above a 3.5. To further verify if the

plant would be suitable for a city was by cross checking if the city had variable temperature and sunshine and if the plant was sensitive to changes in temperature and sunshine. If the plant was not suitable for the variance of the city, it would then be removed from the list. To determine the sensitivity of the plant to temperature and sunshine, we used a multiple regression model from statsmodel.api to find the coefficients of sunshine and temperature on the health score. If the coefficients were higher (i.e., greater correlation to the health score) for a specific climate condition, the plant would be deemed sensitive to that climate condition.

Q2: - Use sunshine data and temperature and precipitation data

To answer this question, we used the merged city climate dataset to visualize the trends of the weather entries in each city to find the correlations between precipitation vs. sunshine and temperature vs. sunshine. We utilized the matplotlib.pyplot library to create the scatter plots with a line of best fit to find the correlations between the metrics.

Q3: - Use the dataset with a multiple regression model to visualize correlations

To answer this question, we further analyzed the plant health data using the same multiple regression model to determine the coefficients of the correlations between temperature and humidity on the health of a plant. Since we wanted to visualize the correlations with a 3D graph, we limited the predictors to only the temperature and humidity of the room that the plant was in. After obtaining the coefficients from the multiple regression model, we then plotted a 3D plane of best fit along with a 3D scatter plot to visualize the relationships. Since this correlation is based on the type of plant, we answered the question for one of the most common houseplants with that being a pothos (*epipremnum aureum*).

EDA Results

In our EDA, we performed several steps to prepare the data for the rest of our analysis. First, we cleaned the weather and plant health datasets by removing irrelevant data and ensuring consistency across variables. We also took time to examine the distribution of plant counts and city counts in the datasets. Initially, we had data for 159 cities, but after merging the datasets and addressing missing values, we were left with 143 cities. The plant dataset contained a diverse set of plant species, each with varying counts, while the city dataset provided a large number of U.S. cities, offering broad geographic coverage for our analysis.

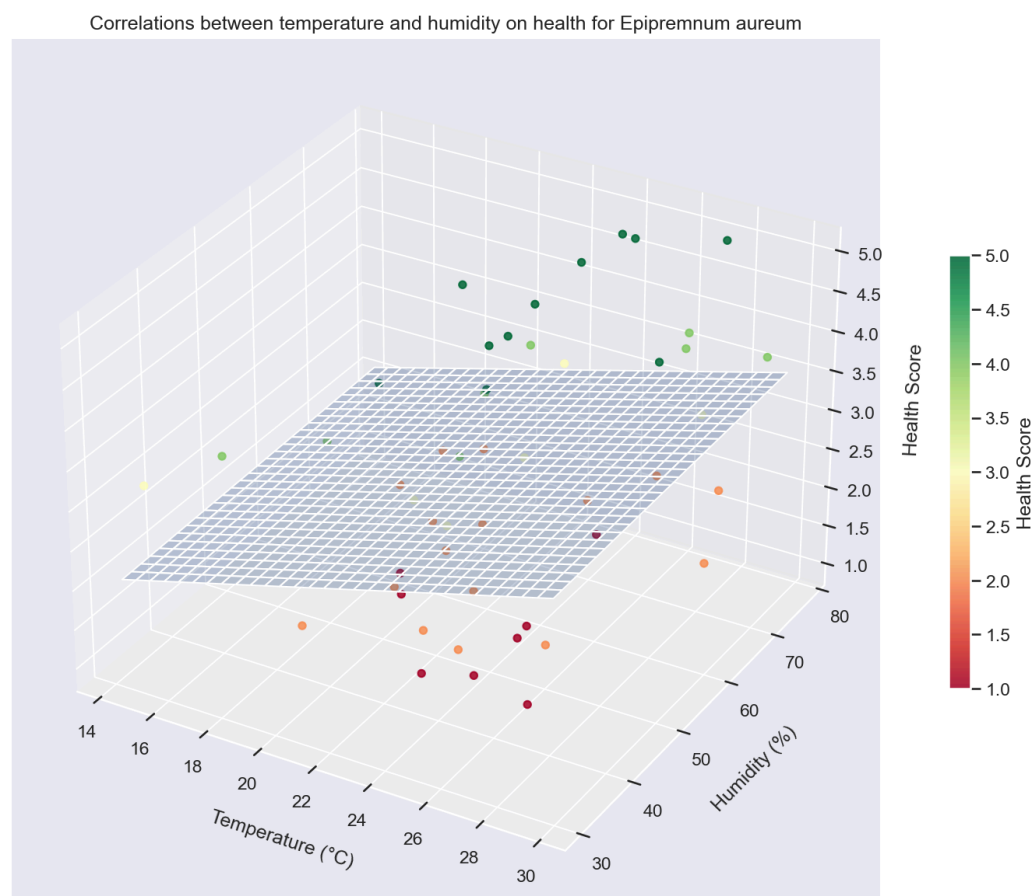
During the EDA, we merged the two datasets based on city, creating a unified dataset that allowed us to analyze environmental factors such as sunshine, temperature, and precipitation in relation to plant health. We visualized the relationships between key

variables, including sunshine and temperature, and examined their correlation with plant health scores. These steps laid the foundation for further analysis, helping us gain key insights into how environmental conditions influence plant health across different regions.

From this process, we learned that cities with higher sunshine percentages tend to have higher temperatures and lower precipitation, which directly influence plant growth. This insight was crucial in understanding the environmental factors that affect plant health. Additionally, the inclusion of plant and city counts provided a better understanding of the dataset's breadth, ensuring we had sufficient data for analysis and that the models we built later in the process would be reliable.

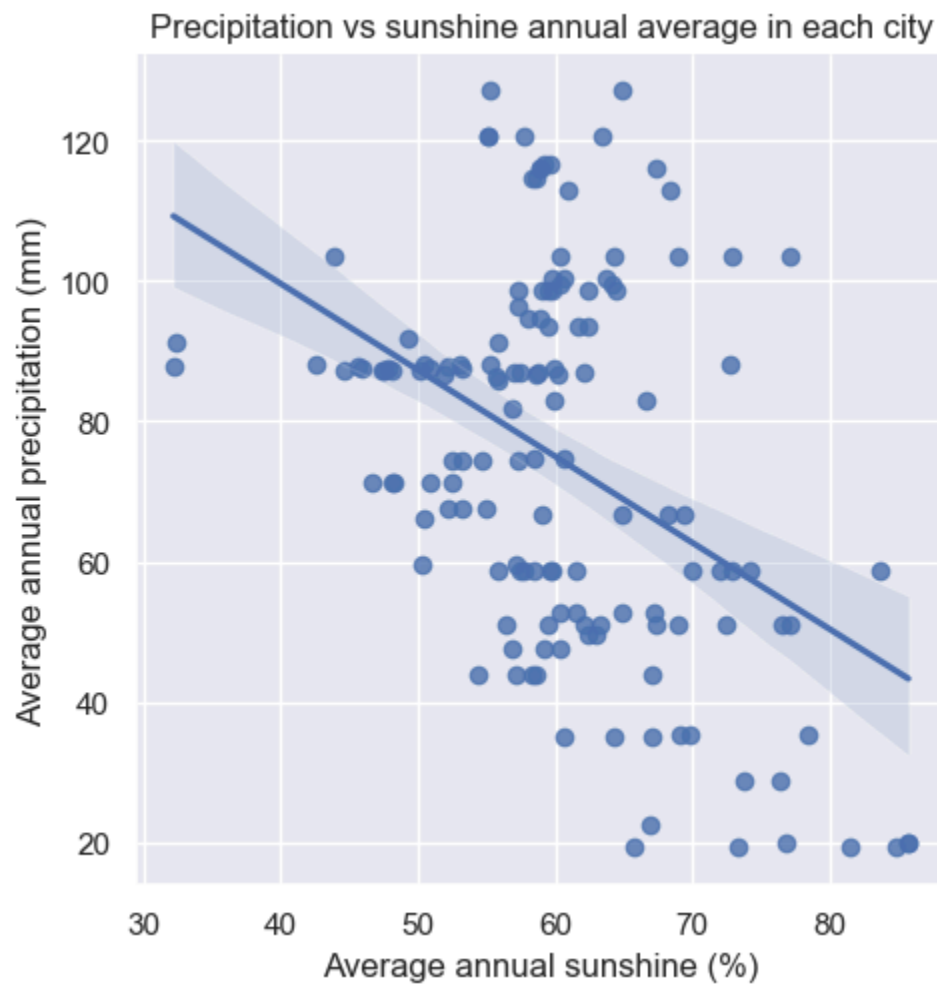
Visualizations:

Plant Data Visualizations

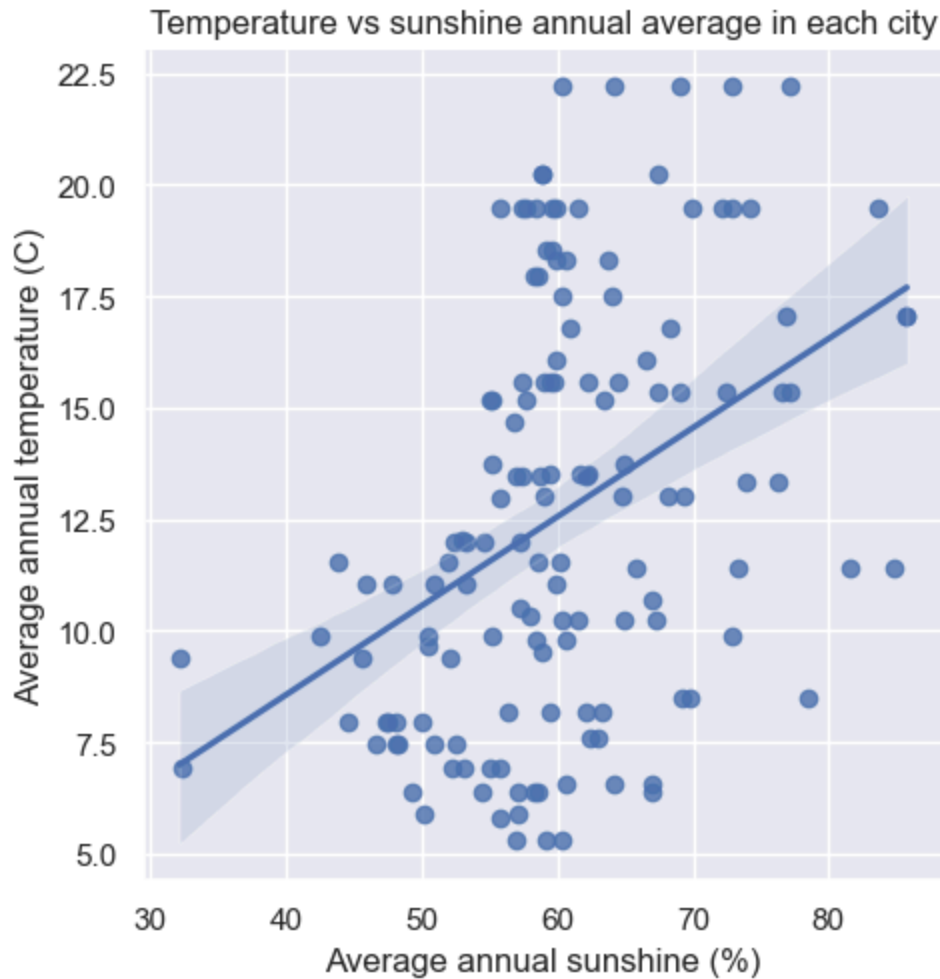


(Figure 1: a 3D representation of the room temperature and humidity's relationship with the health scores for *Epipremnum aureum*)

City Data Visualizations



(Figure 2: A scattered plot with a regression line representing the relationship between average annual sunshine and average annual precipitation.)



(Figure 3: A scattered plot with a regression line representing a relationship between average annual temperature and average annual sunshine.)

Work Plan Evaluation

This was our initial work plan:

We plan to meet every week

1. Data collection and API setup (6 hours).
2. Data cleaning and integration (8 hours).
3. Data analysis (7 hours).
 - Perform correlation analysis and statistical tests to explore relationships between air quality, environmental factors, and plant health.
4. Visualization (8 hours).

- Create additional visualizations such as bar charts, scatter plots, and line graphs for comparing data across cities.

5. Report writing and presentation (2 hours).

We believe our estimates were fairly accurate, as we ended up spending the amount of time we initially projected on the project. Our estimates closely aligned with reality because we accounted for the time needed to consult resources and figure out how to implement unfamiliar tasks, rather than underestimating the effort required.

Results:

1. What plants would be best suited for the Seattle area?

After running the prediction model several times with Seattle's climate, we found that *begonia maculata*, *philodendron hederaceum*, and *Schefflera arboricola* were produced the most, making them the most suitable for the Seattle area.

2. How does the annual percentage of sunshine affect temperature fluctuation and precipitation in different cities?

The graphs presented show clear correlations between sunshine, temperature, and precipitation across different cities. Figure 2 reveals an inverse correlation between sunshine and precipitation. As the percentage of annual sunshine increases, the average monthly precipitation decreases, indicating that cities with more sunshine generally experience lower levels of rainfall. Conversely, Figure 3 illustrates a positive correlation between annual average sunshine and average monthly temperature, indicating that cities with more sunlight tend to experience higher temperatures. This relationship suggests that increased sunshine likely contributes to warmer climates, as evidenced by the upward trend in the data points. These findings highlight the interconnectedness of sunshine, temperature, and precipitation patterns in shaping the climate characteristics of different regions.

3. How do environmental conditions such as temperature and humidity influence plant health and growth indoors?

In our analysis of how environmental conditions, such as temperature and humidity, influence indoor plant health and growth, we examined the correlations between temperature, humidity, and plant health scores. Based on the 3D plot shown in Figure 1, which displays the relationship between temperature, humidity, and the health of *Epipremnum aureum*, we observed that plants in warmer temperatures with optimal

humidity levels tend to have higher health scores. This finding was supported by our multiple regression model and visual data analysis. It is clear from our results that controlling indoor temperature and humidity is essential for promoting healthy plant growth, as these environmental factors directly affect plant health.

Impact and Limitations

There are several important limitations and potential implications to consider.

First, the plant health data was collected in house settings rather than in controlled laboratory environments. This introduces variability, as plants grown indoors can be influenced by a range of factors, including humidity, light sources, and soil conditions, which may not be captured in this dataset. The results may not fully reflect how plants would perform in more consistent or optimized environments, such as greenhouses or experimental labs. The generalizability of the findings may be limited when applying them to different types of indoor plant settings.

Additionally, temperature data was not available for every city in the dataset, so in some instances, we had to generalize temperature information based on state-level data. This method may pose challenges, particularly in larger states or regions with diverse climates, where temperature variations between cities can be significant. This could skew results for cities experiencing unusual weather patterns.

There is also a potential risk of excluding certain plant species or cities that were not adequately represented in the dataset, limiting the applicability of the recommendations for people outside of the studied regions or with uncommon plant preferences.

In terms of potential biases, the reliance on the datasets used in this analysis may introduce inherent biases in which cities or plant species are more thoroughly represented. For instance, regions with more available data may be overrepresented, while areas with sparse climate data might be underrepresented. Additionally, the plant health scores were based on subjective assessments, which could be influenced by personal preferences or environmental factors that were not consistently measured across all cases.

While this analysis offers helpful insights, it should not be used as a definitive guide for plant selection without considering other local factors such as specific care practices and plant variety. Users should treat the conclusions as a general starting point, but they should also take into account their own specific growing conditions and needs before applying the results.

Challenge Goals

The challenge goals outlined in the proposal and the EDA were largely maintained, with slight adjustments made during implementation to better align with the data and analysis process.

1. Utilizing Machine Learning Models:

We initially planned to use a gradient boosting model to predict which plants would be best suited for different environments, such as specific city climates. This goal was fully realized, and the model was successfully implemented. However, we also explored other machine learning techniques, such as linear regression, to compare results and ensure the robustness of our predictions. The decision to focus on gradient boosting was driven by its ability to handle complex, non-linear relationships between environmental variables and plant health, which was a critical aspect of our analysis. This model ultimately provided valuable insights into plant suitability based on temperature, sunshine, and precipitation data.

2. Working with multiple real-world, large datasets:

We also aimed to integrate and work with large, real-world datasets, specifically combining environmental data (temperature, sunshine, precipitation) with indoor plant health information from Kaggle. This aspect of the project was largely successful, although we encountered challenges related to missing or incomplete data, particularly for temperature information in some cities. As a result, we had to generalize temperature data based on state averages in cases where city-specific data was unavailable. Despite these adjustments, the integration of multiple datasets allowed us to create a comprehensive view of the environmental factors influencing plant health and enabled us to draw meaningful conclusions from the combined data.

Overall, the implementation of the challenge goals was successful, with only minor adjustments to the data handling and modeling process. We would say that we expanded our challenge goals. These changes were necessary to address gaps in the data and ensure the analysis remained accurate and reliable.

Testing

To test the methods used to do EDA and the Machine learning preprocessing, we used a mix of assert statements, a smaller csv, and visual checking. For the more quantitative functions like `find_missing_data` and `find_unique_values`, we used assert statements

with a smaller dataset that we could easily predict the results. For the more convoluted and specific methods like the cleaner methods for the datasets, we manually checked the new dataframes by converting them into a viewable .csv file and checking if the unnecessary columns were removed. We tested the helper functions in the preprocessing using assert statements that required more computation. For the formatting used for training and predicting, we made sure that the column names for the altered city climate data and plant data matched using assert statements and visual checks by saving the .csvs so that the model functioned properly.

Collaboration

To help us code and to fix our mistakes, we continually checked StackOverflow for common mistakes when doing data analysis along with the documentation for the different pandas and matplotlib methods and available plots to use in seaborn. We also used those websites for statsmodel.api and XGBoost to determine what functions to use to train the model for our project.